

11/9/26

MOBILE PROGRAMMING II

SOFTWARE ENGINEERING

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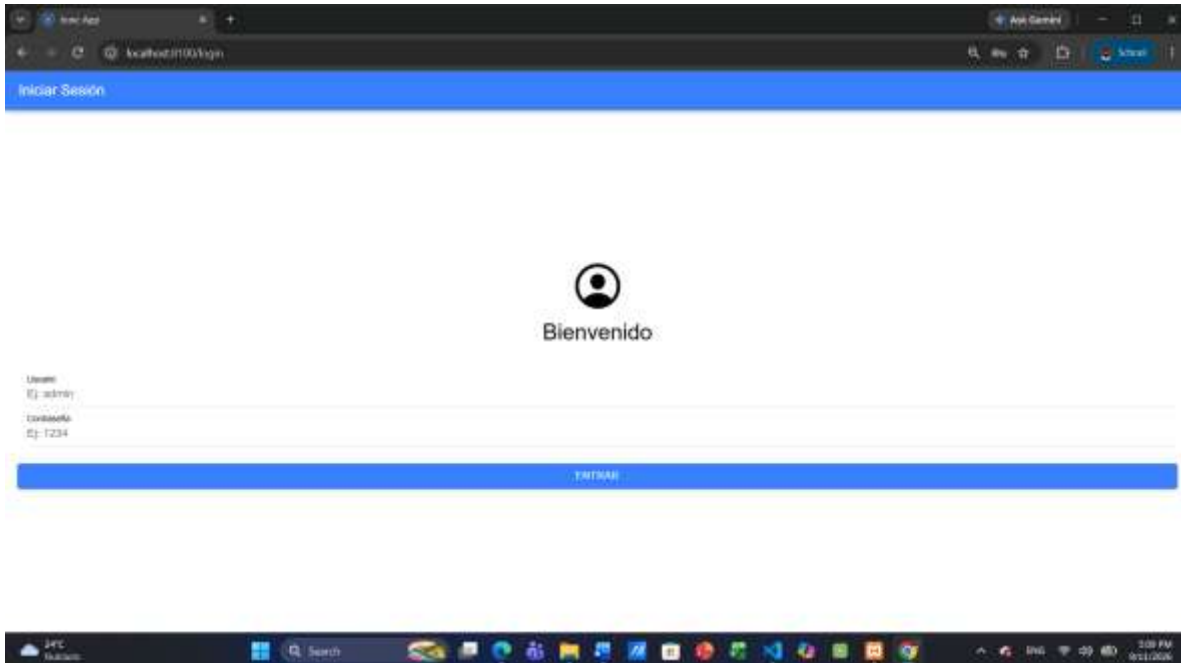
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Creation of the AI-assisted Ionic project

1. Tabs or views

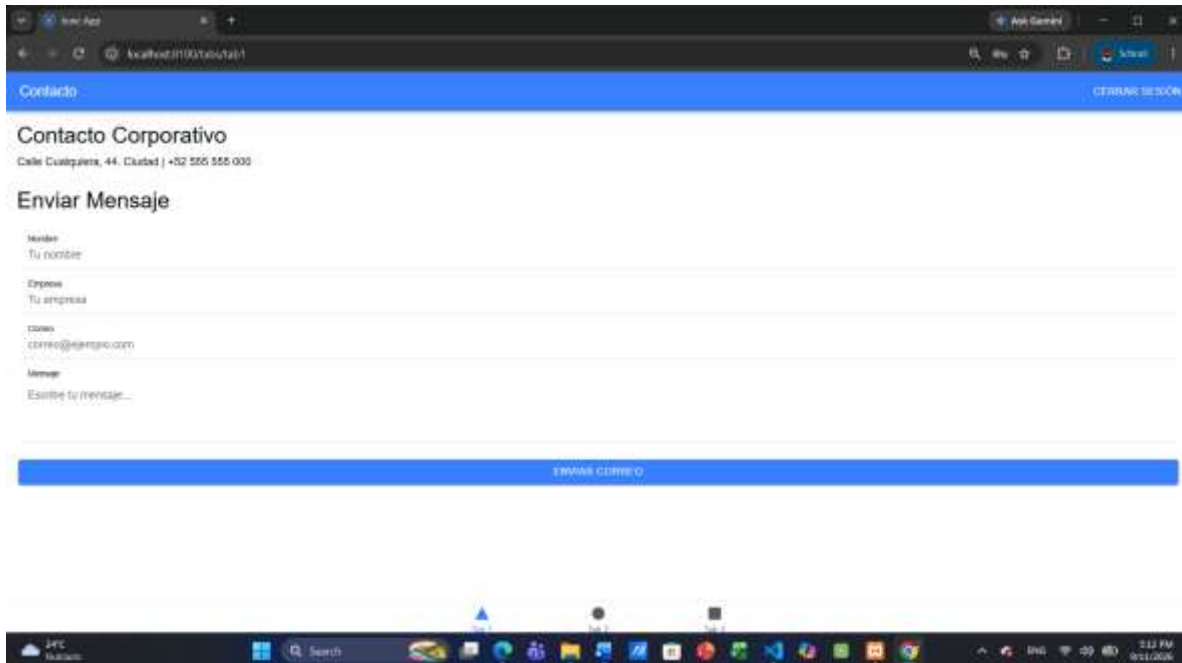
1.1 Login page

Here's the login page, in which I used a similar aspect to match it with the others' views. Use "admin" for user and "1234" for password, it will access.



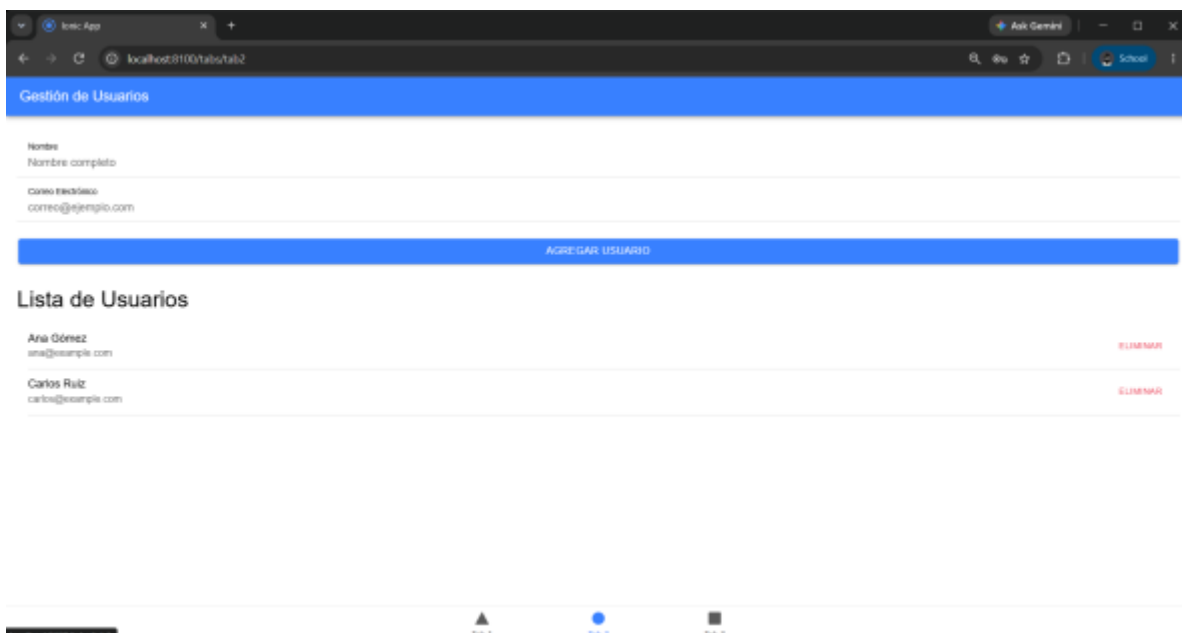
1.2 Contact tab

Here is a contact page in which you can add your personal info and send it. Also there's a logout option available on the upper right corner.



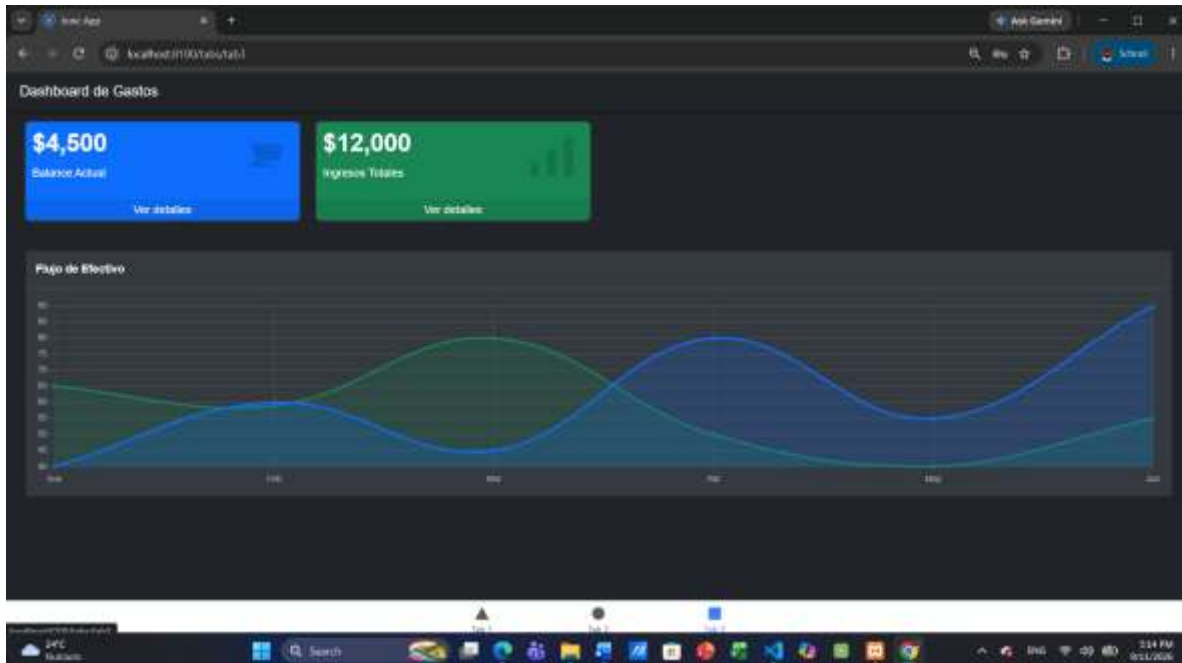
1.3 User management tab

Here's the user's tab, where you can add/delete users. Now I'm still thinking about what to add or edit on this tab.



1.4 Dashboard page

As you recommended me, I used AdminLTE v4 but now I'm also deciding what to exactly include here.



2. App Data Initial Model

2.1 Models' Folder

Inside the folder, there's the model.ts in which it justifies the effective flow that we can see on the graphics. Also it matches with sql.

The screenshot shows a code editor with the 'transactionmodels.ts' file open. The file contains two TypeScript interfaces. The first interface, 'Transaction', is defined as follows:

```
1 export interface Transaction {
2   id: string;
3   tipo: 'ingreso' | 'gasto';
4   monto: number;
5   categoria: string;
6   fecha: Date;
7   descripcion?: string;
8 }
9
```

The second interface, 'ResumenFinanciero', is defined as follows:

```
10 export interface ResumenFinanciero {
11   balanceActual: number;
12   ingresosTotales: number;
13   gastosTotales: number;
14 }
```

The left sidebar of the code editor shows a file explorer with the following structure:

- bank_project_s2
 - node_modules
 - src
 - app
 - explora-container
 - guards
 - login
 - models
 - transactionmodels.ts
 - services
 - api.service.ts
 - tab1
 - tab2
 - tab3
 - tab3-routing.module.ts
 - tab3.module.ts
 - tab3.page.html

3. Execution screenshots.

3.1 Ionic project

As you can see, here the project is running correctly on port 8100.

```

PS C:\PROJECT\ionic_project_v2> ionic serve
[ng] ion-select_3.entry-RYIDSHLD.js | ion-select_3-entry | 60.69 kB |
[ng] ion-input.entry-C2XC6XFQ.js | ion-input-entry | 59.03 kB |
[ng] ion-modal.entry-WNGRHCK6.js | ion-modal-entry | 55.20 kB |
[ng] ion-popover.entry-H4GVG86Q.js | ion-popover-entry | 49.68 kB |
[ng] ion-alert.entry-YQSO3IBM.js | ion-alert-entry | 48.38 kB |
[ng] ion-range.entry-NSLNTPRS.js | ion-range-entry | 45.13 kB |
[ng] ion-refresher_2.entry-6YKIBUSC.js | ion-refresher_2-entry | 41.52 kB |
[ng] chunk-ODGHCPS4.js | - | 38.18 kB |
[ng] ion-segment_2.entry-TDU4DPS4.js | ion-segment_2-entry | 36.53 kB |
[ng] ...and 69 more lazy chunks files. Use "--verbose" to show all the files.
[ng] Application bundle generation complete. [3.450 seconds] - 2026-09-11T23:00:19.314Z
[ng] ▲ [WARNING] The glob pattern import("./**/*.entry.js") did not match any files [empty-glob]
[ng] node_modules/@stencil/core/internal/client/index.js:182:4:
[ng] 182 | ./${bundleId}.entry.js${cacheBustParams ? "?" + cacheBustPar...
[ng]     | ~~~~~
[ng]
[ng] NOTE: Raw file sizes do not reflect development server per-request transformations.
[ng] One or more browsers which are configured in the project's Browserslist configuration fall outside Angular's browser support for this version.
[ng] Unsupported browsers:
[ng] chrome 110, chrome 109, chrome 108, chrome 107, edge 110, edge 109, edge 108, edge 107, firefox
111, firefox 110, firefox 109, firefox 108, firefox 107, firefox 106, ios_saf 16.3, ios_saf 16.2, ios
_saf 16.1, safari 16.3, safari 16.2, safari 16.1

[INFO] Development server running!

Local: http://localhost:8100

Use Ctrl+C to quit this process

[ng] → Local: http://localhost:8100/
[ng] → press h + enter to show help
[INFO] Browser window opened to http://localhost:8100!

```

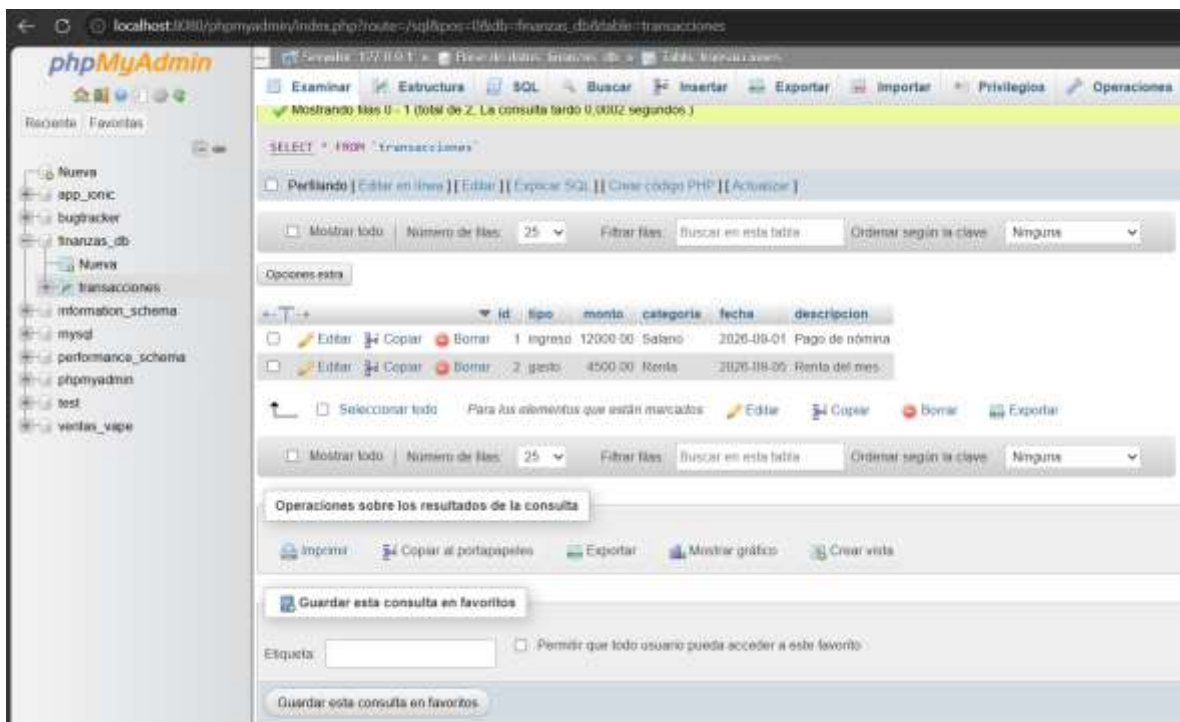
3.2 SQL Connection

Here, you can appreciate the successful connection to SQL Server, the API is working.

```
PS C:\PROJECT> cd ionic_project_v2
PS C:\PROJECT\ionic_project_v2> cd api-backend
PS C:\PROJECT\ionic_project_v2\api-backend> node index.js
Servidor API corriendo en http://localhost:3000
Error al conectar con MySQL: Access denied for user 'admin'@'localhost' (using password: YES)
PS C:\PROJECT\ionic_project_v2\api-backend> node index.js
Servidor API corriendo en http://localhost:3000
Conexión exitosa a MySQL
[]
```

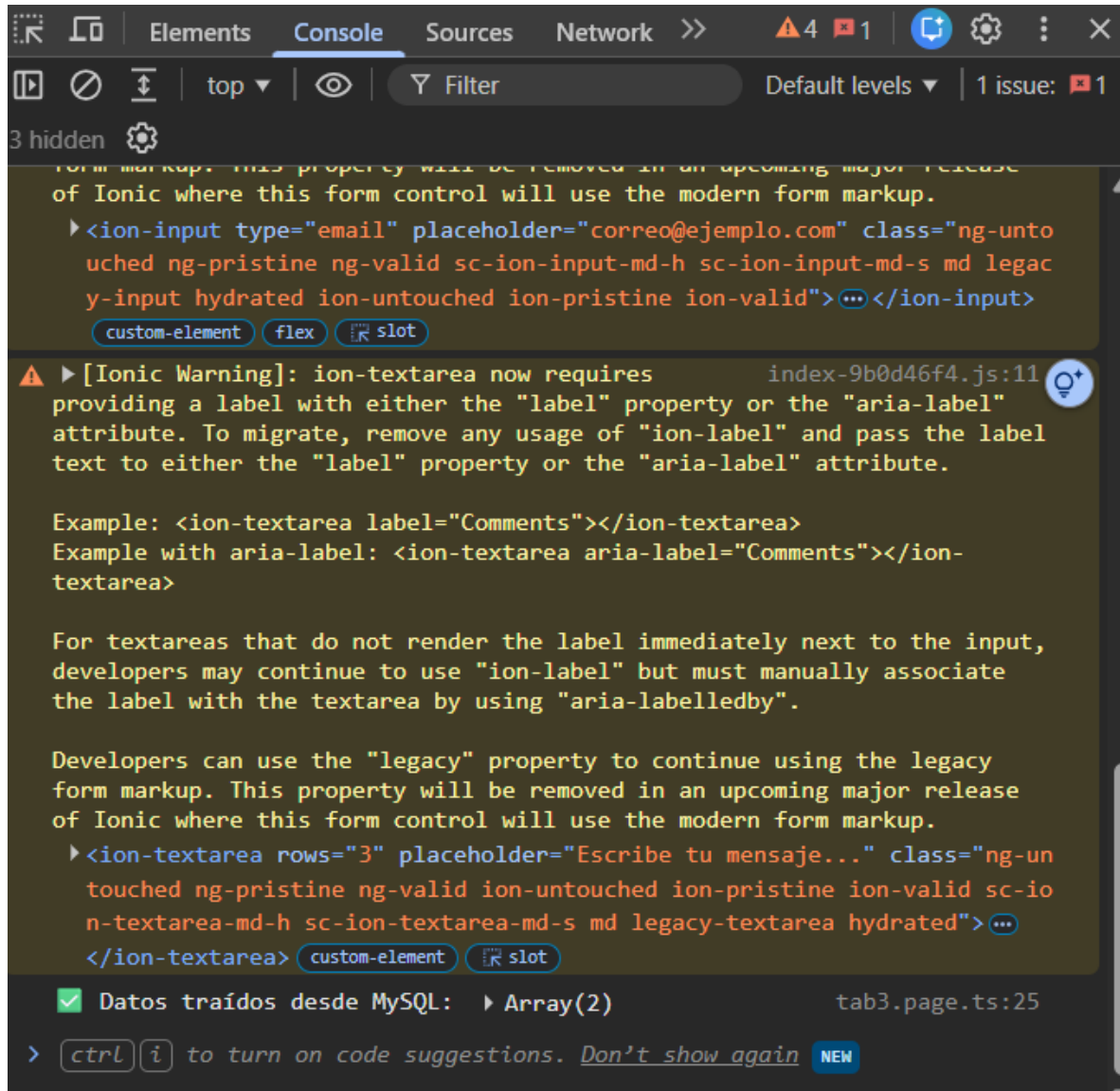
3.3 SQL database

Here, I want to show you the created database that is called “finanzas_db”, it has two categories (Salario, renta)



3.4 Console message

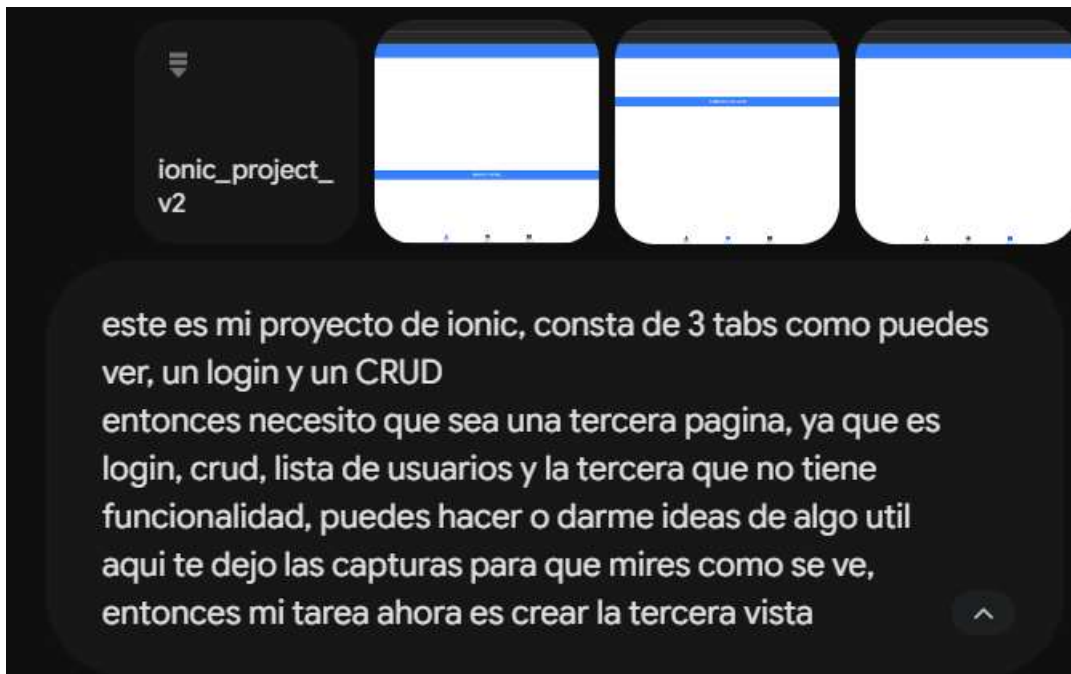
Here, you can see the text “Datos traídos desde MySQL” that proves the API works well.



4. Prompts Used with IA.

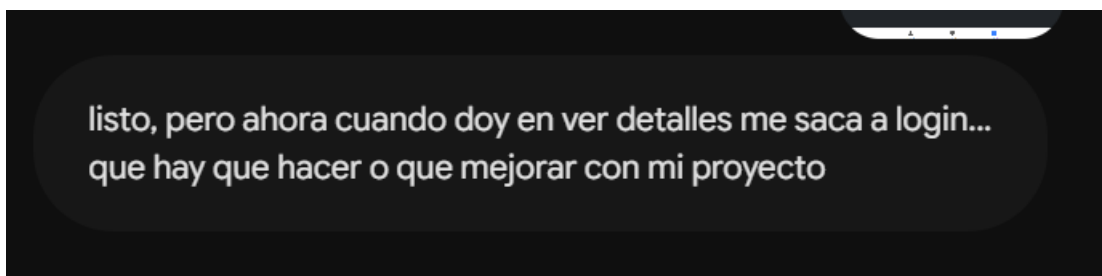
4.1 Starting again...

As I had to create again the project, I asked IA with some help based on what I had after creating my new project.



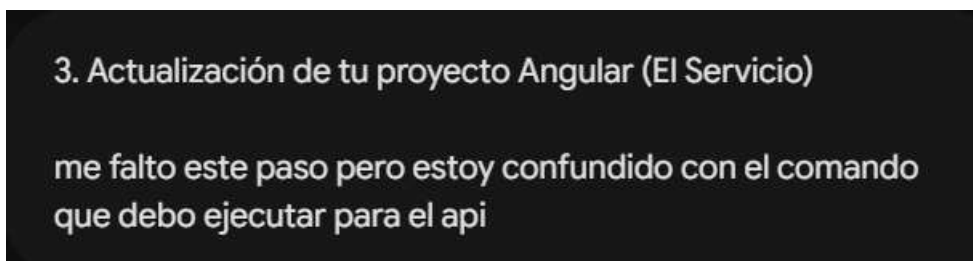
4.2 Correcting errors.

I asked IA to help me with some issues I got with the flow of my project.



4.3 Testing service

Here I got so confused with the commands I had to execute on the terminal, and I did too many changes on different codes, so I asked IA helping me with a little “guide” for testing my service worked correctly.



5. Accepted, modified or denied AI generated code.

5.1 Accepted AI Code

- **Database Script (db.sql):** The CREATE DATABASE statement, transacciones table schema, and initial mock data (INSERT).
- **HTTP Module Setup:** Importing and adding HttpClientModule to app.module.ts.

5.2 Modified AI Code

- **Express API Server (index.js):** Adjusted the mysql.createConnection credentials (changed user from admin to root with an empty password) to match the local XAMPP configuration.
- **Dashboard Component (tab3.page.ts):** Embedded the getTransacciones() subscription inside ionViewDidEnter() while preserving the existing Chart.js initialization logic.
- **TypeScript Interface (transaccion.model.ts):** Tailored the Transaccion model fields to match the database column types strictly.

5.3 Discarded AI code

- **Duplicate Services (data-api.ts & .spec.ts files):** Removed leftover auto-generated files caused by initial CLI naming conflicts to maintain a clean project structure.